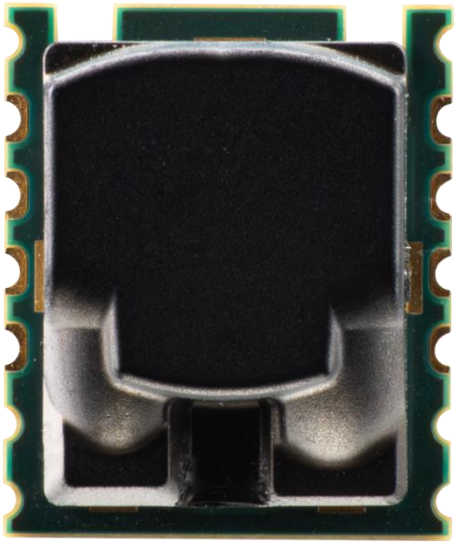


Senseair S12 CO_2

Full performance at a fraction of expected size

Senseair S12 CO_2 boasts the performance and technology of true NDIR CO_2 sensing at a fraction of the expected size. Designed for the next generation of battery, wireless and HVAC applications, Senseair S12 is fully compliant with ANSI/ASHRAE Standard 62.1-2025, RESET grade B, and the WELL Building Standard® (WELL v2™).

Thanks to LED technology, S12 combines ultra-low power consumption with uncompromising accuracy. Its reduced thickness and compact form make it ideal wherever space is at a premium, whether in battery systems, HVAC units, or duct installations, without sacrificing reliability.

Product overview *

Article number	012-0-0001
Operating principle	Non-dispersive infrared (NDIR)
Measured gas	CO_2
Measurement range	0–10 000 ppm
Accuracy	±(30 ppm + 3 % of reading)
Operating conditions	-10–60 °C 0–95 %RH
Power supply	3.0–5.5 V
Average current	< 34 µA
Communication	UART, I^2C
Maintenance	Maintenance free
Life expectancy	> 15 years
Dimensions	18 x 15 x 6 mm

* Preliminary specifications. May be changed without notice.

Key benefits

- SMD reflow solderable
- High precision true NDIR CO_2 Sensing
- Ultra Compact "Micro" Format
- Versatile mounting and placement options
- Very low peak current
- Compliant with building standards:
  - ANSI/ASHRAE Standard 62.1-2025
  - RESET grade B
  - WELL Building Standard® (WELL v2™)
  - Title 24, California Energy Code

Applications

- Battery powered applications
- Indoor air quality equipment
- Building management systems
- Demand controlled ventilation systems
- Agricultural applications

Copyright © 2026 Senseair AB

1. Sensor performance *

1.1 Sensing performance

Parameter	Conditions	Value
Target gas	-	Carbon dioxide (CO_2)
Operating principle	-	Non-dispersive infrared (NDIR)
Gas sampling method	-	Diffusion
Measurement range [1]	Standard	0–5000 ppm
	Extended	5001–10 000 ppm
Accuracy [2,3,4]	ANSI/ASHRAE environments [5]	±(30 ppm + 3% of reading)
	0–5000 ppm	±(50 ppm + 3% of reading)
	5001–10 000 ppm	±(70 ppm + 3% of reading) or ±10% of reading – whichever is greater
Resolution	-	1 ppm
Measurement interval	Default	2 s; Configurable from 2 s
Sampling	Default	59 samples; Configurable
Response time	$T_{63\%}$	< 30 s
Compensation [6]	Temperature	On-board sensor element
	Pressure	Pressure value must be provided by host system. Otherwise, dependency is 1.6% of reading per kPa deviation from normal pressure

* Preliminary specifications. May be changed without notice.

1. Sensor is designed to measure in the range of 400–5000 ppm (extended range up to 10 000 ppm). Exposure to concentrations below 400 ppm may affect performance. For measurements of concentrations below 400 ppm, ABC shall be turned off.
2. Accuracy in the table is defined at 25 °C, 1013 mbar ambient pressure and 50% RH. The accuracy for the complete operating conditions is specified in chapter 4.2. Specification is referenced to uncertainty of calibration gas mixtures ±1 %.
3. Shipping, rough handling and assembly can temporarily affect the accuracy of the sensor. Accuracy can be fully restored by forced recalibration or after a maximum of 3 ABC periods after mounting the sensor.
4. Continuous mode, filtered concentration readings.
5. S12 CO_2 complies with ANSI/ASHRAE Standard 62.1-2025, concentrations 600, 1000, and 2500 ppm measured at sea level at 25 °C.
6. CO_2 readings are temperature compensated. Optional host system can provide an ambient pressure value for pressure compensated CO_2 readings. The S12 does not have an integrated pressure sensor.

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

1.2 General performance

Parameter	Conditions	Value
Operating temperature [1]		-10–60 °C
Operating humidity	Non-condensing	0–95% RH
Power supply		3.05–5.5 V
Peak current [2]		< 20 mA
Average Current [3]	Default settings	< 34 µA
Nominal dimensions	(L x W x H)	18 x 15 x 6 mm
Life expectancy		> 15 years
Storage temperature		-40–85 °C
Weight		< 3 g
Communication interface		UART / I²C
Maintenance [4]		Maintenance-free

1. Sensor is calibrated to meet product specifications within standard range 15–35 °C, extended range -10–60 °C. Sensor can survive temperatures outside of operating range, but measurement accuracy is not guaranteed.
2. To guarantee the functionality of the sensor, the voltage supply must be kept, and the maximum current must be considered.
3. The average current consumption depends on configured measurement period and number of samples per measurement.
4. For maintenance-free operation, ABC (Automatic Baseline Correction) must be enabled.

2. Pin configuration and functions

Pin #	Symbol	I/O Type	Description
1	GND	Power	Ground
2	VBB	Power	Supply voltage
3	RxD / SDA	I/O	UART receive input / I²C bidirectional serial data.
4	TxD / SCL	I/O	UART transmit output / I²C clock input.
5	EN	Input	Enable (active high). Drive pin high to power up sensor. High = Sensor enabled Low = Shutdown mode Do not leave floating. Can be connected to VBB if sensor used in continuous mode.
6	DE	Output	Communication direction pin. Connect to transceivers direction pin. High when S12 is transmitting. Leave floating if not used.
7	nRDY	Output	Measurement ready output. Open drain output. Can be changed to push-pull output by register setting.
8	COMSEL	Input	Communication select High = UART (Internal pull-up, can be left floating) Low = I²C (Connect to GND)

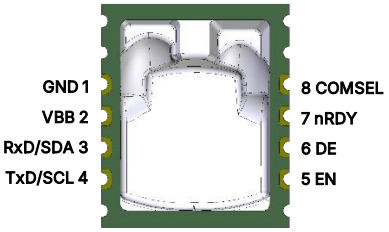


Figure 1. Pin configuration of Senseair S12 CO_2

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

3. Drawing - Dimensions

Dimensional drawing of sensor with dimensions and their tolerances in millimetres.

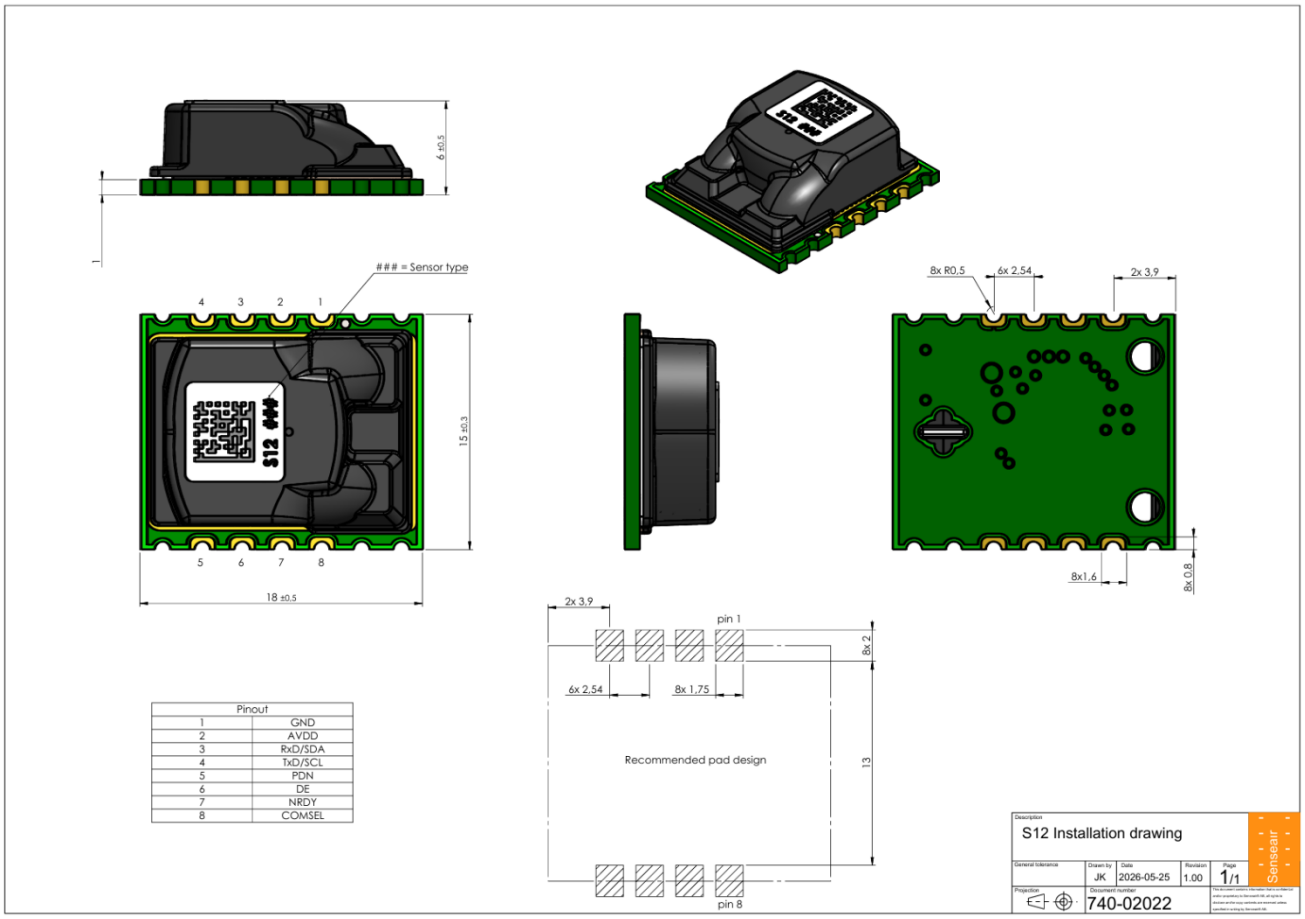


Figure 2. Dimensions of Senseair S12 CO_2

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

4. Specifications

Over operating temperature range, unless otherwise noted.

4.1 Recommended operating conditions

Symbol	Description	Min	Typ	Max	Unit
VBB	Supply voltage	3.05	3.3	5.5	V
COMSEL	Communication select	-0.3	-	VBB+0.3	V
EN	Enable	-0.3	-	VBB+0.3	V
RxD/SDA	UART / I²C	-0.3	-	VBB+0.3	V
TxD/SCL	UART / I²C	-0.3	-	VBB+0.3	V

Parameter	Description	Symbol	Min	Typ	Max	Unit
Power supply rise time [1, 2]	Time until VBB is set to the operating voltage from 0.2 V.	PSUP	1 µ		10 m	s
Power-on reset time [1, 2]	Time until ready for communication after PSUP.	PORT			30	ms
Shutdown voltage [2, 3, 5]	Shutdown voltage for POR re-starting.	SDV			0.2	V
Power supply interval time [1, 2, 3]	Voltage retention time below SDV for POR re-starting.	PSINT	10			ms
Capacitor charge time [4]	Time until internal capacitor are charged	CCT	128			ms

1. Values of these parameters are not guaranteed and not tested in production. Therefore, for applications that require high reliability, please do not use Power-On Reset (POR) but instead use the PDN pin to do reset.
2. POR circuit detects rising edge of VBB, resets the internal circuit, and initializes the registers. The host system must take into account PSUP and PORT times to communicate with the S12.
3. Unless this condition is satisfied, reset may not be correctly performed.

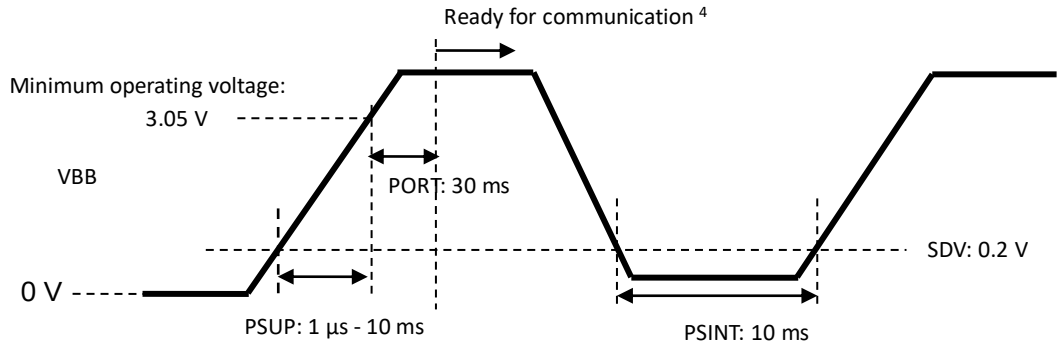


Figure 3: Power supply conditions

4. In single measurement mode, writing 1 to the Single Measurement Start register during internal capacitor charging time(CCT) it will be ignored.

5. The EEPROM is not protected against power loss. Do not power off the sensor during an EEPROM write operation. Verify that the nRDY pin indicates completion before turning off the sensor.

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

4.2 Sensor accuracy over operating range

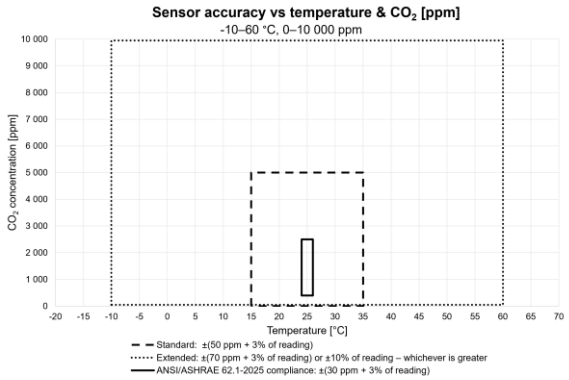


Figure 4: Accuracy over temperature and CO_2

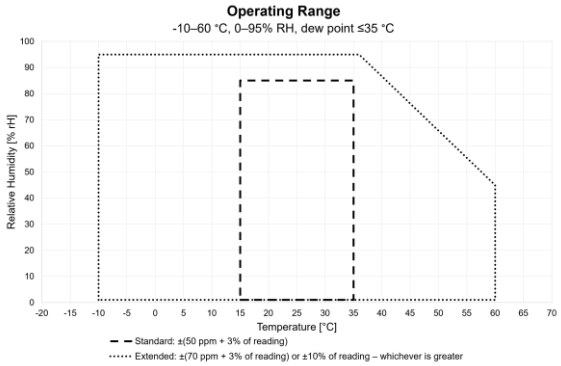


Figure 5: Operating Range

4.3 Electrical Characteristics

Parameter	Description	Symbol	Min	Max	Unit
High level input voltage	RxD/SDA, Tx/SCL, EN, COMSEL	VIH	70%VBB		V
Low level input voltage	RxD/SDA, Tx/SCL, EN, COMSEL	VIL		30%VBB	V

4.4 Absolute Maximum Ratings

Parameter	Description	Min	Max	Unit
Input current	RxD/SDA, Tx/SCL, EN, COMSEL	-15	15	mA

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

4.5 EEPROM Characteristics

The EEPROM is not protected against power loss. Do not power off the sensor during an EEPROM write operation. Verify that the nRDY pin indicates completion (+275ms for single measurement mode) before turning off the sensor.

Do not write to reserved registers. Writing to reserved registers may cause unexpected behaviour and impact the sensor performance.

Parameter	Min	Typ	Max	Unit
Retention	15			years
Endurance [1]			100 000	cycles
Writing time [2]			180	ms
Waiting time after nRDY pin changes status during single measurement mode			275	ms
Ready for communication after enable pin-high (PORT)		30		ms
Start single measurement command after ready for communication (in single measurement mode) (CCT)	128			ms

1. Values of these parameters are not guaranteed and not tested in production.
2. When writing to the following EEPROM mapped holding registers all at once or individually. The writing time applies for both single and multiple registers writing operation.

Measurement Mode	ABC Target
Measurement Period	Static IIR filter parameter
Number of samples	Meter control
ABC period	MB / I^2C address

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

4.6 Measurement cycle timing diagram

The following diagrams do not include the EEPROM write operation that occurs when the host system writes to the EE-mapping holding registers.

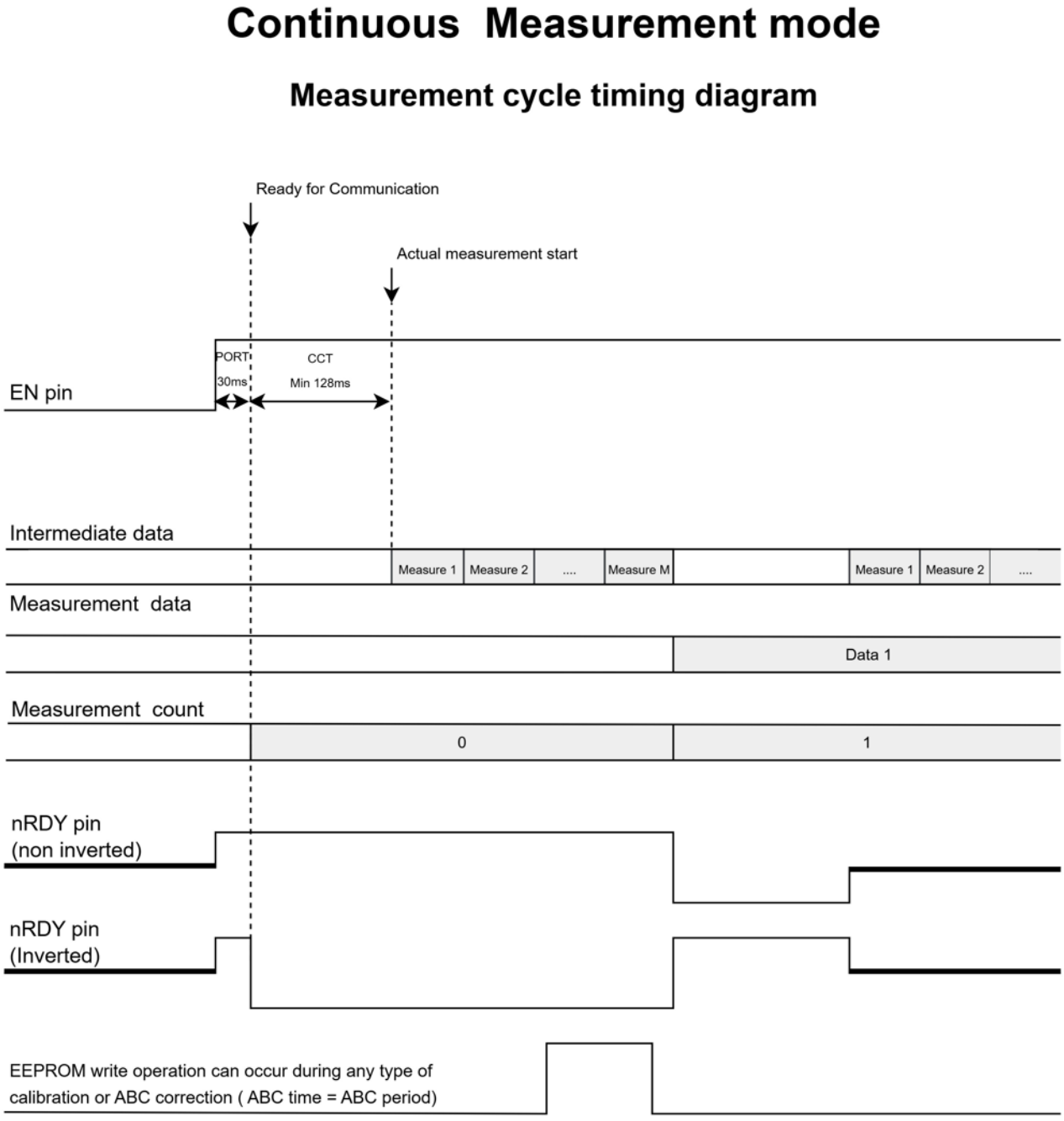


Figure 6: Continuous measurement mode, Measurement cycle timing

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

Single Measurement mode

Measurement cycle timing diagram

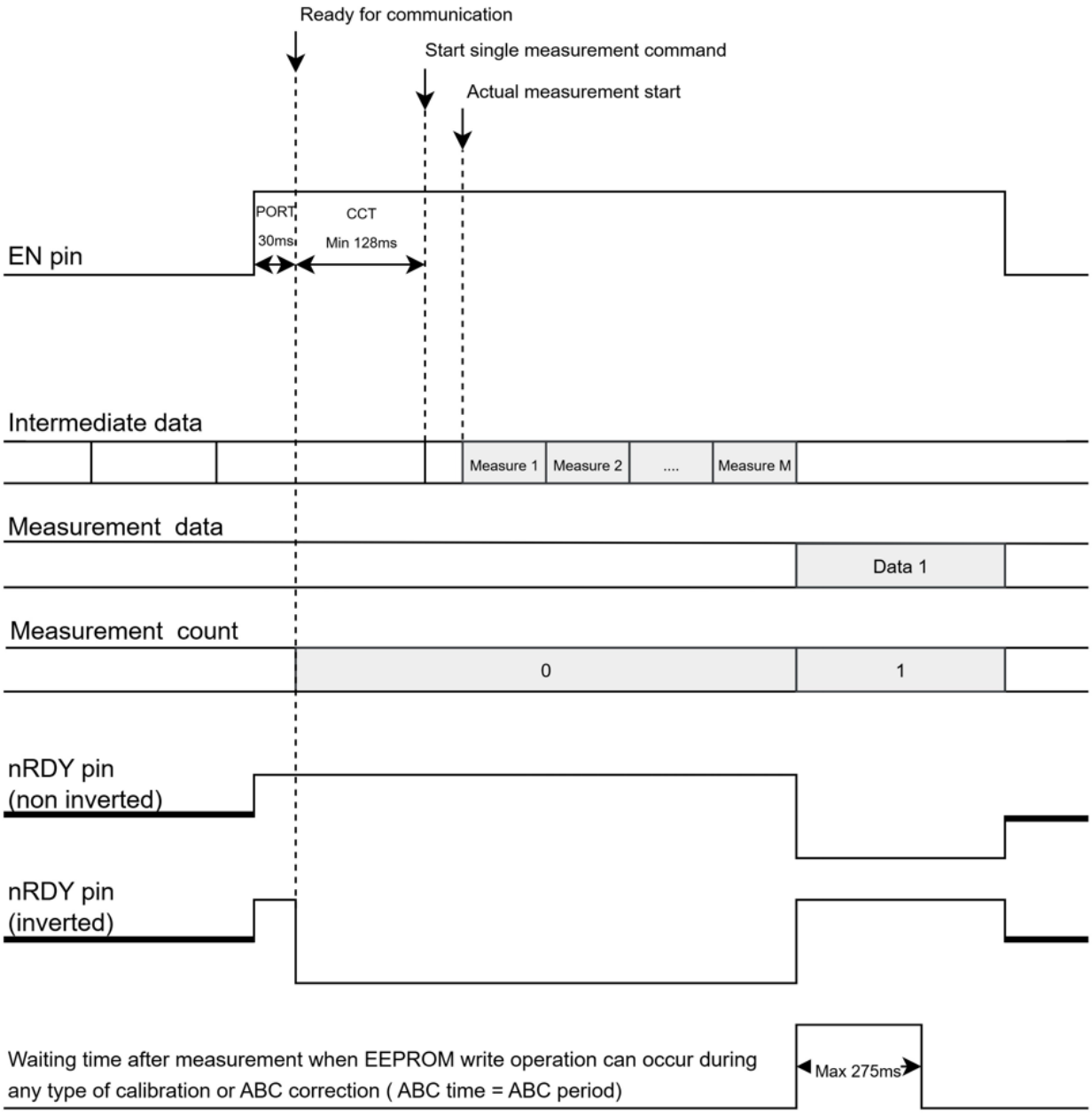


Figure 7: Single measurement mode, Measurement cycle timing

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

5. Sensor documentation

5.1 Measurement mode

The S12 CO_2 supports two modes of operation:

1. Continuous measurement mode
2. Single measurement mode

The **default** operation mode for S12 CO_2 is **Continuous measurement mode**.

1) In Continuous measurement mode, the sensor measures at regular intervals (measurement period, default setting 2 s). The host can read measurement data after each measurement and does not need to send any command to trigger measurements.

2) In Single measurement mode, the sensor waits for the hosts command to measure. The host needs to send a command sequence to trigger each measurement. By using this function, the current consumption can be further optimized and gives more flexibility how often the sensor should start a measurement without changing basic settings of the sensor.

5.2 Communication

See “Modbus on S12 CO_2” (TDE15927) and “I^2C on S12 CO_2” (TDE15933).

5.3 Maintenance

Senseair S12 CO_2 has a built-in self-correcting ABC algorithm**.** ABC is enabled by default, and ABC period is adjustable by host. Discuss your application with Senseair, in order to get advice for a proper calibration strategy.

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

5.4 Revision History

Date	Revision	Section	Changes
September 2025	1	All	Initial release of document
January 2026	2	All	Removal of "Micro" from sensor name. Update of operating range from 0–85% RH to 0–95% RH. Update of standard measurement range from 400–5000 ppm to 0–5000 ppm. Addition of "±(70 ppm + 3% of reading)" to extended range accuracy. Update of default sampling settings, from 64 to 59 samples. Update of weight from <5 g to <3 g. Addition of "standard range 15–35 °C, extended range -10–60 °C" to Note 1 in 1.2 General performance. Update of Figure 3: Accuracy over temperature and CO2 Figure 4: Operating Range. Removal of fields that were missing information.
May 2026	3	1st page	Updated header and footer of Marketing Page. Changed height from 7 to 6 mm.
July 2026	4	1st page	Updated text on Marketing page.
		2, 4.1	Changed pin naming, VDD → VBB, RDYN → nRDY.
		2	Updated description of pins 5, 6 and 7
		3	Updated drawing with tolerances ad more information about the Sensor ID label placement. Adding extra dimensions.
		4.1	Updated text in note 4&5 in Figure 3.
		4.3 & 4.4	Added sections "Electrical Characteristics" and "Absolute Maximum Ratings".
		4.5	Updated the complete clause
		4.6	Added section 4.6 "Measuring cycle timing" for both Single and continuous measurement mode.

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB

IMPORTANT NOTICES

1. Senseair reserves the right to make changes to the information contained in this document without notice. When you consider any use or application of Senseair product stipulated in this document ("Product"), please make inquiries the sales office of Senseair or authorised distributors as to current status of the Products.
2. All information included in this document are provided only to illustrate the operation and application examples of Senseair Products. Senseair neither makes warranties or representations with respect to the accuracy or completeness of the information contained in this document nor grants any license to any intellectual property rights or any other rights of Senseair or any third party with respect to the information in this document. You are fully responsible for use of such information contained in this document in your product design or applications. Senseair ASSUMES NO LIABILITY FOR ANY LOSSES INCURRED BY YOU OR THIRD PARTIES ARISING FROM THE USE OF SUCH INFORMATION IN YOUR PRODUCT DESIGN OR APPLICATIONS.
3. The Product is neither intended nor warranted for use in equipment or systems that require extraordinarily high levels of quality and/or reliability and/or a malfunction or failure of which may cause loss of human life, bodily injury, serious property damage or serious public impact, including but not limited to, equipment used in nuclear facilities, equipment used in the aerospace industry, medical equipment, equipment used for automobiles, trains, ships and other transportation, traffic signalling equipment, equipment used to control combustions or explosions, safety devices, elevators and escalators, devices related to electric power, and equipment used in finance-related fields. Do not use Product for the above use unless specifically agreed by Senseair in writing.
4. Though Senseair works continually to improve the Product's quality and reliability, you are responsible for complying with safety standards and for providing adequate designs and safeguards for your hardware, software and systems which minimise risk and avoid situations in which a malfunction or failure of the Product could cause loss of human life, bodily injury or damage to property, including data loss or corruption.
5. Do not use or otherwise make available the Product or related technology or any information contained in this document for any military purposes, including without limitation, for the design, development, use, stockpiling or manufacturing of nuclear, chemical, or biological weapons or missile technology products (mass destruction weapons). When exporting the Products or related technology or any information contained in this document, you should comply with the applicable export control laws and regulations and follow the procedures required by such laws and regulations. The Products and related technology may not be used for or incorporated into any products or systems whose manufacture, use, or sale is prohibited under any applicable domestic or foreign laws or regulations.
6. Please contact Senseair sales representative for details as to environmental matters such as the RoHS compatibility of the Product. Please use the Product in compliance with all applicable laws and regulations that regulate the inclusion or use of controlled substances, including without limitation, the EU RoHS Directive. Senseair assumes no liability for damages or losses occurring as a result of noncompliance with applicable laws and regulations.
7. Resale of the Product with provisions different from the statement and/or technical features set forth in this document shall immediately void any warranty granted by Senseair for the Product and shall not create or extend in any manner whatsoever, any liability of Senseair.
8. This document may not be reproduced or duplicated, in any form, in whole or in part, without prior written consent of Senseair.

For regular updates, check the website: https://senseair.com/ Copyright © 2026 Senseair AB